Confidence-Shielded Evaluator-Conditioned Fusion for Risk-Controlled Infrared–Visible Object Detection

Jianyong Zhao[a], Zigao Wu[a,*], Mengfei Ma[a]

[a] *North China Electric Power University, Hebei, Baoding, 071003, China*

Abstract

Infrared–visible fusion is often treated as a global preprocessing gain, yet the same fused image may improve one detector and degrade another. We introduce Confidence-Shielded Evaluator-Conditioned Fusion (CS-ECF), a label-free policy that scores visible, consensus-detail fusion (CDF), PIAFusion, and SPGFusion candidates with the target detector. CS-ECF averages the five highest confidences of each candidate and selects a fused input only when its score reaches 1.5 times that of visible; otherwise, visible is retained. The margin was frozen before testing.

Across fixed-split LLVIP, grouped six-class M3FD, and official DroneVehicle test subsets, comprising 12 confirmatory checkpoints, nine checkpoint point estimates improved over visible, although no individual checkpoint contrast survived Holm correction and the sign test was not significant ($p = 0.073$). The dataset-balanced AP50 gain was 0.0133 with a training-run-by-group 95% interval of [0.0026, 0.0261]. Against Always CDF, the balanced difference was 0.0102 [-0.00057, 0.02980], so superiority over this fixed input was not established. Removing the shield increased the worst AP50 loss from 0.0040 to 0.0584. Two zero-shot evaluators gained 0.021–0.025 AP50 over visible without separating from CDF. A third evaluator, fine-tuned on LLVIP, lost 0.0108 AP50 relative to visible; its scene-group interval crossed zero, although shielding improved the unshielded policy by 0.0153. This negative check shows that confidence arbitration is not universally portable. The evidence supports bounded, evaluator-dependent risk control, not uniform improvement, CDF dominance, or low-latency deployment. Serial candidate generation and four detector evaluations required approximately 0.71 s per image pair.

Keywords: infrared–visible image fusion, object detection, evaluator-conditioned fusion, confidence calibration, risk control, thermal imaging

1. Introduction

Infrared–visible imaging combines the spatial detail and color of visible sensors with thermal evidence that can remain available when illumination is poor, smoke reduces contrast, or targets have weak visual texture. This complementarity has motivated pixel-, feature-, and network-based fusion methods for surveillance, autonomous perception, and thermal imaging systems. [1–10] In detection-oriented applications, the value of a fused image is not determined by visual appeal alone; it is the change in detector outputs under a specified operating condition.

Current practice often selects one fused representation and evaluates it with one detector on one benchmark. This design assumes that a globally preferred fusion result remains preferable across detector architectures and training seeds. The assumption is fragile because detectors differ in receptive fields, confidence calibration, training data, and sensitivity to high-frequency detail

*Corresponding author.

Email addresses: shaohaojia2001@163.com (Jianyong Zhao), zgwu@ncepu.edu.cn (Zigao Wu), MYmmf01@163.com (Mengfei Ma)

URL: https://orcid.org/0009-0005-3251-2785 (Jianyong Zhao), https://orcid.org/0000-0002-7449-343X (Zigao Wu), https://orcid.org/0009-0007-1328-1314 (Mengfei Ma)

[11, 12]. Consequently, a fusion rule that improves one evaluator may reduce the average precision (AP) of another on the same image.

Per-image selection offers a natural response: run several fusion candidates and choose the one with the strongest detector response. Yet unconstrained selection converts small calibration differences into frequent input switches and can produce severe losses when a spurious high-confidence response is mistaken for improved detection. The problem is therefore not only to predict which fused image looks best, but to decide when the evidence is strong enough to depart from the visible-image safety baseline, a decision related to the classical reject option in recognition [13].

Recent infrared–visible fusion and detection studies have optimized shared fusion networks, cross-modal feature fusion, or hierarchical detectors [14–17]. General-purpose detectors provide the evaluator side of this problem [18, 19]. These studies establish the relevance of joint fusion and detection, but they generally train or optimize a detector-fusion system. They do not establish a label-free arbitration rule whose switching threshold is fixed independently of the confirmatory tests and whose behavior is reported at the level of detector checkpoint, image, and scene or frame group.

We formulate infrared–visible fusion selection as evaluator-conditioned input arbitration. CS-ECF evaluates four candidates with the target detector, computes a top-five confidence score, and accepts a fused candidate only when its score reaches 1.5 times that of the visible input. The threshold is selected on validation data and frozen before testing. We evaluate the policy on LLVIP, M3FD, and DroneVehicle with model-, image-, and group-aware uncertainty, compare it with fixed fusion and an unshielded selector, and report the cost and failure boundaries. The contributions are:

1. an evaluator-conditioned formulation of fusion-input selection that does not use test labels;
2. a confidence shield that reduces catastrophic switching while retaining positive dataset-balanced gains;
3. a three-dataset confirmation that separates image-level uncertainty, scene/group uncertainty, and checkpoint variability, including explicit evidence that the M3FD gain is concentrated in the people class, the absence of a separated advantage over Always CDF, and a fine-tuned-evaluator counterexample to universal portability.

2. Related work

2.1. Fusion representations

Classical infrared–visible fusion methods combine complementary evidence in a hand-designed transform or representation. Representative examples use multi-scale top-hat transforms, non-subsampled contourlet transforms with compressed sensing, two-scale saliency decomposition, and three-scale framelet or shear representations [3, 20–22]. Sparse-representation methods instead describe local image content with learned or overcomplete dictionaries before applying a coefficient-level fusion rule [23, 24]. These approaches make target contrast, edges, and local saliency explicit. Their combining rules are typically designed for image-level visual quality, however, rather than for the ranking that a particular detector assigns to the resulting image.

Learned fusion models replace handcrafted combination rules with trainable feature extraction and reconstruction. DenseFuse, IFCNN, U2Fusion, and RFN-Nest are representative encoder–decoder or residual designs [4, 6–8]. Adversarial formulations add discriminative losses for texture and target preservation [5, 25–27]. Dual-path wavelet networks and transformer-based designs subsequently added multi-scale context and long-range interaction [28–31], while diffusion models introduced generative priors for fusion [9, 10]. These methods enlarge the candidate design space, but they generally define a single fused output. They do not specify how to choose among several available candidates for a fixed deployed detector.

2.2. Task-aware fusion and detection

Task-aware fusion introduces an illumination, semantic, or saliency signal into the transfer rule. PIAFusion conditions fusion on illumination [32]; SPGFusion and MTFusion incorporate semantic priors [33, 34]; and classification-saliency or masked autoencoder designs use auxiliary representations to decide which features to retain [35, 36]. Such priors make the transfer more selective, but a proxy such as illumination or segmentation does not guarantee a better confidence ordering for every detector.

On the detection side, one-stage and DETR-style architectures provide different confidence behaviour and receptive-field structure [18, 19, 37, 38]. Recent infrared–visible systems integrate fusion with detection through shared backbones, cross-modal feature exchange, saliency masking, or hierarchical fusion [14–17, 39, 40]. Joint optimization can align the fused representation with the detector, but the resulting coupling is learned for a specified training objective. It does not by itself provide a label-free rule for selecting among already generated fusion candidates when the target checkpoint or data composition changes.

2.3. Evaluation and selection boundary

Fusion quality has commonly been assessed with reference or no-reference image metrics [1, 2]. Those scores summarise source retention or perceptual quality; they do not identify which of several inputs will produce the higher AP for a particular detector. Multispectral detection studies show that thermal evidence can improve robustness under poor illumination [41, 42], but gains depend on architecture, training data, and the operating regime [11, 12]. The same dependence is visible across the detector checkpoints used in this study.

The decision to depart from a visible-image fallback resembles the classical reject option [13], but the object of the decision is an input candidate rather than a class prediction. The threshold must therefore be specified separately from accuracy, fixed before confirmation, and evaluated against both average gain and worst-case loss. Reporting only a fused-image score or a single benchmark pairing leaves that decision boundary untested.

2.4. Positioning

CS-ECF does not introduce another image-fusion operator. It treats an existing fusion library as a set of alternative inputs and uses the target detector as a label-free evaluator. Relative to task-aware networks, it requires neither a new fusion model nor joint detector-fusion training. Relative to fixed image-quality rankings, it measures the quantity of interest directly and separates detector checkpoints, scene or frame groups, and independent evaluators. The contribution is therefore an arbitration policy plus a risk-boundary evaluation, not a claim that any one fused representation is universally superior.

3. Methods

3.1. Problem formulation

Let x_v and x_i denote a registered visible–infrared pair, and let $\mathcal{D}$ denote the target detector checkpoint. A candidate $c \in \mathcal{C}$ produces an input image x_c, where $\mathcal{C} = \{\mathrm{vis}, \mathrm{cdf}, \mathrm{pia}, \mathrm{spg}\}$. The policy returns one candidate per pair without access to ground-truth boxes:

$$g(x_v, x_i; \mathcal{D}) \in \mathcal{C}. \tag{1}$$

The final detection output is $\mathcal{D}(x_{g(x_v,x_i;\mathcal{D})})$. The detector is therefore reused for candidate evaluation and final inference; it is not an external label oracle.

3.2. Fusion candidates

The visible input is always included as a safety candidate. CDF combines a pixel-mean image and an IFCNN image through sign-consistent high-frequency details. Let x_m denote the pixel-mean image, let x_f denote the IFCNN image, and let G_σ denote Gaussian blur with $\sigma = 5$. The corresponding detail images are $d_m = x_m - G_\sigma * x_m$ and $d_i = x_f - G_\sigma * x_f$. Their consensus detail is

$$d(x) = \begin{cases} \operatorname{sign}(d_m + d_i)\min(|d_m|, |d_i|), & d_m d_i > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

A thermal-validity mask M is formed from the infrared top-hat response, infrared intensity, visible luminance, connected components, and smoothing. The top-hat threshold $q_{\mathrm{TH}} = 0.95$ is the response quantile above which a pixel is considered thermally salient; it is distinct from the candidate score q_c defined below. We used a 31×31 elliptical top-hat kernel, an infrared floor of 120, visible luminance below its 75th percentile, 5×5 morphological opening, and retention of connected components with at least 64 pixels. The surviving mask was dilated and smoothed with 31×31 kernels. The shared detail is injected into the LAB luminance channel:

$$L' = L + \alpha M d. \tag{3}$$

The frozen settings are $q_{\mathrm{TH}} = 0.95$, $\alpha = 0.20$, and $\sigma = 5$. PIAFusion and SPGFusion are generated with their official model definitions and pretrained weights under the same image-pair protocol. [4, 32, 33]

3.3. Confidence-shielded arbitration

For candidate c, let $s_{c,1} \geq s_{c,2} \geq \cdots$ be detector confidences. We define

$$q_c = \frac{1}{\min(K, n_c)} \sum_{j=1}^{\min(K,n_c)} s_{c,j}, \qquad K = 5, \tag{4}$$

where n_c is the number of detections. Candidates with no detections receive $q_c = -1$. The highest-scoring candidate is $c^* = \arg\max_c q_c$, with ties resolved in favor of visible. The selected input is

$$c = \begin{cases} c^*, & c^* \neq \text{vis and } q_{c^*} \geq 1.5 q_{\mathrm{vis}}, \\ \text{vis}, & \text{otherwise.} \end{cases} \tag{5}$$

The four-candidate library and the top-five score were fixed before the confirmatory tests. The score averages five boxes rather than relying on one maximum, while retaining a small set that is not dominated by the low-confidence tail. The candidate library was chosen to span visible, region-selected, learned, and published fusion families. We evaluated margins $\{1.00, 1.10, 1.20, 1.30, 1.50, 1.75, 2.00\}$ on the 150 LLVIP validation images and 400 M3FD validation images using two detectors per dataset, before scoring the confirmatory test images. The objective was to maximize the dataset-balanced validation gain subject to no evaluator losing more than 0.0005 AP50. Within this fixed top-five rule, 1.5 was the first setting satisfying the constraint and gave the largest balanced validation gain. It switched 4.7% of validation decisions, compared with 11.4% at 1.2. Table 1 reports a post-freeze sensitivity check over other score depths and margins. The expanded grid was not used to reselect the policy after test performance was observed. The complete pre-freeze search is reported in the Supplementary Material.

The validation gains are selection diagnostics over four evaluator cells. They are not estimates of the 12-checkpoint confirmatory effect and should not be compared directly with the test-set point estimate. The margin was not tuned on DroneVehicle, and its public validation partition was not used for policy selection. DroneVehicle therefore serves as an external domain-transfer check.

Table 1: Validation sensitivity of score depth and visible margin. Entries are dataset-balanced AP50 gain (worst evaluator gain) over the four LLVIP and M3FD validation evaluators. The frozen policy is $K = 5$ and margin 1.5. The expanded grid is descriptive and was not used to reselect the policy after the confirmatory tests.

Margin	$K = 1$	$K = 5$	$K = 10$
1.2	-0.0014 (-0.0048)	-0.0002 (-0.0056)	-0.0029 (-0.0073)
1.5	-0.0000 (-0.0004)	**+0.0008 (-0.0001)**	+0.0004 (-0.0010)
2.0	-0.0000 (-0.0004)	+0.0001 (-0.0004)	+0.0016 (+0.0000)
2.5	0.0000 (0.0000)	+0.0001 (0.0000)	+0.0000 (-0.0004)

Table 2: Training, validation and confirmatory test partitions. The DroneVehicle counts are not reconstructed locally; its checkpoints use the public training and validation partitions recorded by the released model recipe. The evaluated images come only from the official test archive.

Dataset	Train	Validation	Test	Evaluation groups
LLVIP	1,350	150	300	7 scene groups
M3FD	1,600	400	200	2 frame groups
DroneVehicle	public train	public validation	300	3 frame groups

3.4. Datasets and evaluators

The official LLVIP partitions are larger than the experiment subsets [43]. The 300-image test subset contains the first 300 pairs after a uniformly random shuffle (seed 0) of official test pairs with non-empty labels. The release used here contains 3,463 test pairs, all with non-empty labels; 3,163 eligible pairs were therefore not evaluated. The 1,500-image training subset comprises the lexicographically first 1,500 official training pairs with non-empty labels. Within this ordered subset, the first 1,350 pairs form training and the remaining 150 form validation. Train/test, validation/test and train/validation stem intersections are all empty; the three partitions contain 4,479, 419 and 744 pedestrian boxes. The eight LLVIP confirmatory checkpoints are YOLO11n trained on visible images, a modality-agnostic YOLO11n control, YOLO26n seeds 0–2, RT-DETR-l, and YOLOv8n seeds 0 and 1. YOLO11n seed 0 is used for each YOLO11n family; four independently trained runs of each family are retained as a seed sensitivity analysis. All best checkpoints were selected using the 150 validation images, not the scored test subset.

M3FD contributes a grouped six-class split of 1,600 training, 400 validation and 200 test images [44]. Frames are partitioned by consecutive 100-frame group using a fixed group seed, so adjacent frames do not cross splits. The two frame groups in the test set are specified in the released manifest. One YOLO26n and one YOLOv8n visible-trained six-class evaluator were selected on the validation groups.

DroneVehicle contributes 300 image pairs from three fixed consecutive 100-frame groups of the official test archive [45]. The groups were selected with a predefined seed of 2037 before evaluation. The YOLO11x-OBB best and last checkpoints come from one 150-epoch training run (batch size 32, 640-pixel input, seed 0). Because these checkpoints share training data, they are treated as one model cluster rather than independent architectures. The source archive has 4,430,343,899 bytes and SHA-256 `94BF47E493E7A57FD5D900439F614F339E7B6C7181E4D781BCFEBAEB963FFD8E` to identify the exact release used for the test groups.

External evaluator stress tests. We also applied the frozen selector to three checkpoints that were not used to construct the policy or select its margin. YOLOv9t and YOLOv10s were COCO-pretrained models without LLVIP fine-tuning, and only their person-class detections were retained [37, 38]. A third YOLOv9t was initialized from the same COCO weights and fine-tuned for a pre-specified 40 epochs on the 1,350 LLVIP visible training images. Training used AdamW, a learning rate of 0.002 with three-epoch warm-up and cosine decay, weight decay 5×10^{-4}, batch size 16, 640-pixel inputs, horizontal flipping, and brightness/contrast jitter. A reproducible PIL-based

Table 3: Evaluator inventory. "Checkpoints" counts only the confirmatory models used in the policy comparison; additional independent runs are retained for sensitivity analysis.

Dataset	Evaluator	Training input	Seed / checkpoint	Checkpoints
LLVIP	YOLO11n	visible	seed 0	1
LLVIP	YOLO11n	visible + infrared	seed 0	1
LLVIP	YOLO26n	visible	seeds 0, 1, 2	3
LLVIP	RT-DETR-l	visible	seed 0	1
LLVIP	YOLOv8n	visible	seeds 0, 1	2
M3FD	YOLO26n	visible, six classes	seed 0	1
M3FD	YOLOv8n	visible, six classes	seed 0	1
Drone-Vehicle	YOLO11xOBB	VIS, vehicle OBB	best/last, one run	2

loader and the Ultralytics detection loss were used. The last checkpoint was fixed after epoch 40; no epoch was selected using test results.

Detection evaluation. Predictions use a 640-pixel inference size and a confidence floor of 0.001. For YOLO models, non-maximum suppression uses an IoU threshold of 0.7. At each class and each IoU threshold from 0.50 to 0.95 in steps of 0.05, predicted boxes are assigned to ground-truth boxes by one-to-one maximum-cardinality matching over valid pairs. The resulting true-positive flags and confidence scores are pooled across images within a checkpoint and integrated over the 101-point interpolated precision-recall curve using Ultralytics 8.4.189 [46]. AP50 is the mean over the classes present in the dataset; AP50–95 further averages the ten IoU thresholds.

3.5. Statistical analysis

Point estimates use pooled AP50 after concatenating detections across test images within a checkpoint. Image bootstrap intervals resample images within a checkpoint. Group bootstrap intervals resample scenes or consecutive frame groups and are the primary clustered intervals. A detector checkpoint is the unit for architecture-level summaries; images and scenes/frame groups are repeated observations within a checkpoint and are never treated as independent training runs. Checkpoints from one training run remain one run cluster. The dataset-balanced estimand gives LLVIP, M3FD, and DroneVehicle equal weight. M3FD contains only two test groups and LLVIP seven scenes, so its group intervals are explicitly finite-cluster summaries.

The confirmatory multiplicity family contains the 12 prespecified CS-ECF-minus-visible checkpoint contrasts. We used 2,000 paired scene/frame-group bootstrap resamples per checkpoint, computed two-sided bootstrap p-values, and applied Holm correction at family-wise $\alpha = 0.05$; Benjamini–Hochberg q values are also reported. The number of positive checkpoints is additionally summarised with an exact sign test and is not treated as a substitute for effect sizes. The dataset-balanced estimate uses 10,000 run-cluster-by-group resamples; 20 independent 5,000-draw reruns were used to check interval stability. The CS-ECF-versus-Always CDF dataset-balanced contrast is one prespecified secondary estimand; its per-dataset values are descriptive rather than a separate confirmatory family. The three external-evaluator contrasts are an exploratory stress-test family, with Benjamini–Hochberg correction across those three checks.

We define detection-level ECE using ten fixed confidence bins and also report the Brier score for the external evaluators. These calibration quantities are descriptive summaries of matched detections, not calibrated class probabilities. Confidence selection scores use the top-five mean before any calibration adjustment.

Table 4: Confirmatory checkpoint summary and policy comparison. Mean is the unweighted average over the 12 confirmatory checkpoints. Dataset-balanced assigns equal weight to LLVIP, M3FD, and DroneVehicle after averaging within each dataset; worst is the lowest checkpoint difference relative to visible.

Policy	Positive	Mean	Dataset-balanced	Worst
Always CDF	10/12	+0.0052	+0.0031	-0.0059
Always PIAFusion	3/12	-0.1087	-0.0710	-0.2910
Always SPGFusion	2/12	-0.1136	-0.0602	-0.2501
Top-5 no shield	7/12	+0.0132	+0.0332	-0.0584
CS-ECF	9/12	+0.0074	+0.0133	-0.0040

The training-run-by-group bootstrap used the parameters above with `numpy.random.default_rng(2219)` in Python 3.10.22 and NumPy 2.2.6. Spearman association tests used SciPy 1.15.3. The Supplementary Material provides the complete split counts, all per-checkpoint CS-ECF and Always CDF results, the validation-policy record, and the multiplicity, baseline, class, calibration, and thermal analyses. The public analysis script reconstructs each checkpoint and rejects it if its AP does not agree with the published aggregate table before computing the paired interval.

3.6. Implementation and cost

At a standardized fusion input of 384×512, FP32 and batch size one on an NVIDIA RTX 2080 Ti, the CDF pipeline required 22.3 ms, PIAFusion 37.7 ms, and SPGFusion 516.3 ms. Four serial YOLO11x-OBB evaluations at an input size of 640 required 130.0 ms, giving an estimated serial total of 706.3 ms. This estimate excludes model loading or swapping, data loading, and final detection post-processing. SPGFusion's additional peak activation memory was 1453.5 MiB; per-stage memory values must not be summed as an end-to-end peak.

4. Results

4.1. Fusion utility is evaluator-dependent

Across the 12 confirmatory checkpoints, the same candidate library produced both positive and negative gains (Fig. 1). CS-ECF improved nine checkpoint point estimates. Its dataset-balanced AP50 gain was 0.0133, with a training-run-by-group hierarchical 95% interval of [0.0026, 0.0261]. The exact sign test for 9/12 positive contrasts gave $p = 0.073$. After Holm correction across the 12 checkpoint family, no individual contrast had adjusted $p < 0.05$; the smallest raw p-value, for LLVIP/YOLO11n, was 0.005 (Holm-adjusted $p = 0.060$). We therefore interpret the balanced estimate as a small average effect and do not claim that any individual checkpoint improved significantly. The unshielded selector improved seven of 12 checkpoints, had a larger dataset-balanced mean of 0.0332, and produced a worst loss of 0.0584.

4.2. Dataset-specific behavior

On LLVIP, the mean gain was 0.0016 and five of eight checkpoints improved; the model and image intervals crossed zero. On M3FD, both six-class checkpoints improved, with a mean gain of 0.0111. On the independent DroneVehicle test subset, the best and last OBB checkpoints improved by 0.0251 and 0.0293; the combined group interval was [0.0002, 0.0406]. All three DroneVehicle groups were non-negative, but one gained only 0.0002. The two OBB checkpoints originated from the same training run and are not independent architectures.

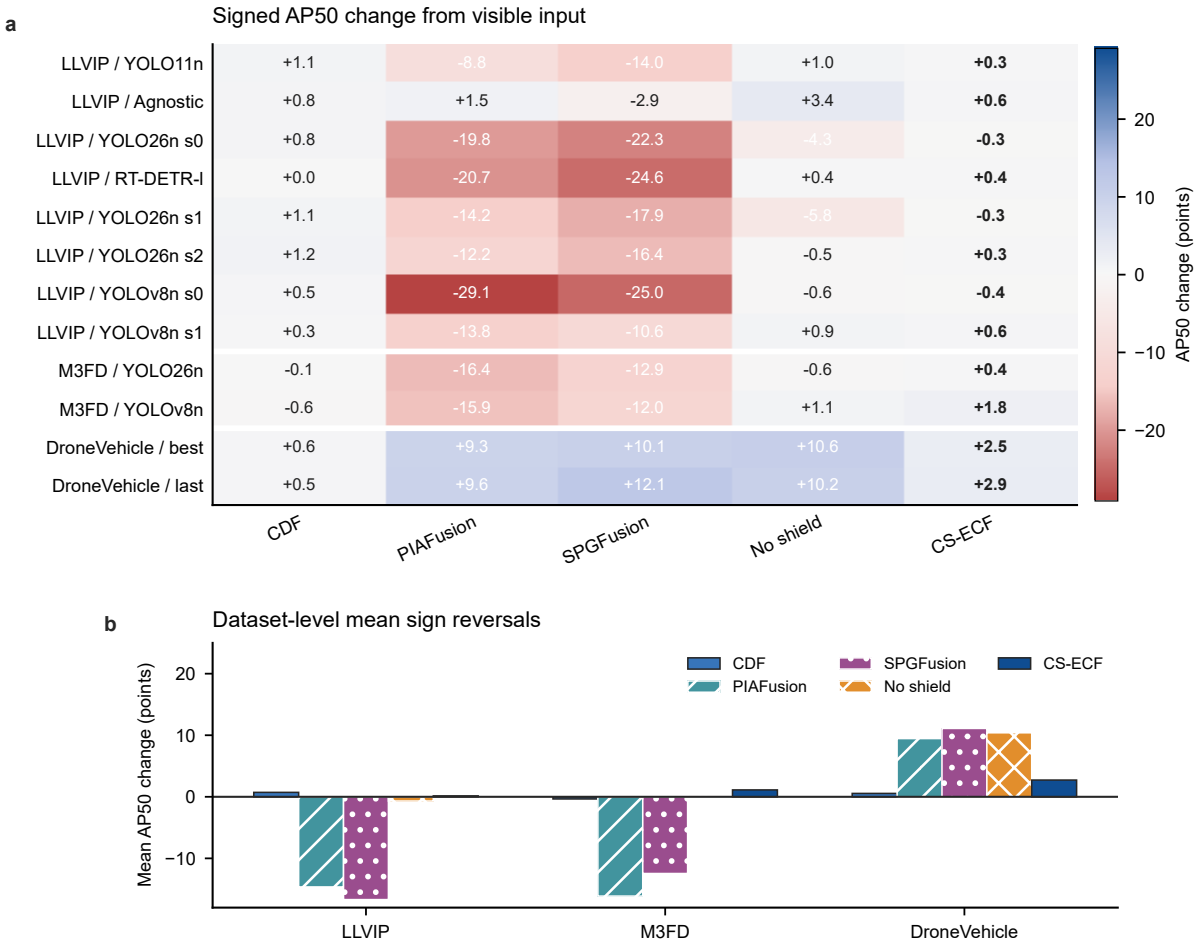


Figure 1: **Evaluator-dependent utility of infrared–visible fusion candidates.** **a**, Signed AP50 change from the visible input for CDF, PIAFusion, SPGFusion, the unshielded selector, and CS-ECF across 12 detector checkpoints. Cells are point contrasts. **b**, Dataset-level mean AP50 changes. Equal candidate sets produce opposite signs across datasets. Source data are provided in `figure1_source.csv`.

Table 5: Paired AP50 difference between CS-ECF and Always CDF. Intervals resample scene or frame groups; the combined interval also resamples training runs within each dataset and then weights datasets equally.

Scope	CS-ECF − Always CDF	95% group-aware interval
LLVIP	−0.0057	[−0.0189, +0.0156]
M3FD	+0.0147	[+0.0100, +0.0159]
DroneVehicle	+0.0217	[−0.0007, +0.0738]
Dataset-balanced	+0.0102	[−0.00057, +0.02980]

4.3. Direct comparison with the strongest fixed baseline

Always CDF is the strongest fixed policy, so it is the relevant test of whether evaluator-conditioned arbitration adds value. Across the 12 checkpoints, CS-ECF exceeded Always CDF in six and fell below it in six. The dataset-balanced difference was 0.0102 AP50 in favour of CS-ECF. Its run-cluster by group interval was $[-0.00057, 0.02980]$ and included zero: LLVIP favoured Always CDF, DroneVehicle was represented by a single training-run cluster, and only M3FD had a clearly positive point contrast. Table 5 therefore reports the dataset-specific contrasts rather than only the pooled endpoint.

The direct comparison changes the claim boundary. CS-ECF is not a statistically separated improvement over Always CDF; its average difference is concentrated in the mixed-class and vehicle-focused evaluations and is negative on LLVIP. The defensible contribution is risk-controlled evaluator-conditioned arbitration and its observed trade-off, not dominance over the strongest fixed input.

4.4. Controls and headroom

We added label-free controls and a labelled upper bound to determine whether the confidence signal adds information beyond choosing a visually strong candidate. The VIF selector chose the fused candidate with the highest mean reference VIF to visible and infrared, and the SSIM selector used the analogous mean structural similarity. Both selected among the three fused candidates only, whereas the entropy selector chose among all four inputs. All image statistics were computed on 256×256 grayscale inputs. Uniform random selection sampled the four candidates independently for each image and checkpoint over 500 allocations. The oracle selected the candidate with the highest per-image AP50 using test labels and is an upper bound only. Images without ground-truth boxes received an AP50 of zero for every candidate, so ties resolved to visible.

The common 256×256 grayscale representation made the image-quality inputs comparable across source resolutions and detector architectures. These selectors used no target labels.

The maximum-reference-VIF selector produced a dataset-balanced gain of 0.0008 with 8/12 positive checkpoints, far below the confidence-based policy. SSIM and entropy selection were strongly negative. CS-ECF lies about 40% of the distance from the mean uniform-random policy to the per-image oracle on dataset-balanced AP50, showing both that detector confidence adds value beyond image-only statistics and that substantial headroom remains.

4.5. External evaluator stress tests

To test whether the result was tied to the evaluator families used during policy development, we first applied the frozen selector to the two COCO-pretrained YOLOv9t and YOLOv10s checkpoints without LLVIP fine-tuning. We then tested the same selector with an LLVIP-fine-tuned YOLOv9t, whose lower test-set performance and larger train-to-test shift provide a stress case for the confidence score.

Both zero-shot evaluators improved relative to visible, while neither provided a separated advantage over Always CDF. The fine-tuned YOLOv9t changed the sign: CS-ECF was 0.0108 AP50 below visible, with a scene-group interval crossing zero, and 0.0040 below Always CDF. The shield nevertheless improved the unshielded selector by 0.0153 AP50, with a group interval

Table 6: Additional policy controls. Dataset-balanced AP50 and worst-checkpoint changes are relative to visible. Positive counts refer to checkpoint-level point estimates. The random row is the mean of 500 uniform candidate allocations per checkpoint; its worst entry is the lowest checkpoint mean. The oracle uses test labels and is an upper bound, not a deployable selector.

Policy	Dataset-balanced AP50	Worst checkpoint	Positive
Always CDF	+0.0031	-0.0059	10/12
Uniform random	-0.0341	-0.1341	2/12
Maximum reference VIF	+0.0008	-0.0059	8/12
Maximum SSIM	-0.0946	-0.2855	3/12
Maximum entropy	-0.0599	-0.2432	2/12
CS-ECF	+0.0133	-0.0040	9/12
Per-image oracle	+0.0833	+0.0205	12/12

Table 7: External evaluator stress tests on the same 300 LLVIP test pairs. Values in the second column are visible / Always CDF / no-shield / CS-ECF AP50. The first two evaluators are COCO-pretrained without LLVIP fine-tuning; the third is fine-tuned on LLVIP and tests a negative portability boundary. Intervals resample LLVIP scene groups. NS, no shield.

Evaluator	AP50 (V/CDF/NS/CS)	CS-ECF − visible	CS-ECF − CDF
YOLOv9t, zero-shot	.7698 / .7888 / .8216 / .7948	$+.0250$ $[+.0053, +.0523]$	$+.0060$ $[-.0277, +.0374]$
YOLOv10s, zero-shot	.8344 / .8617 / .8699 / .8557	$+.0213$ $[+.0041, +.0439]$	$-.0059$ $[-.0266, +.0180]$
YOLOv9t, LLVIP-trained	.6666 / .6598 / .6405 / .6558	$-.0108$ $[-.0342, +.0050]$	$-.0040$ $[-.0206, +.0108]$

of $[+0.0072, +0.0267]$. It selected visible for 230 of 300 images and reduced the worst scene loss from 0.1208 to 0.0724, but did not eliminate the loss.

The shield's tail effect was mixed even among the zero-shot checks. On YOLOv10s it reduced the worst scene loss from 0.0337 to 0.0038; on zero-shot YOLOv9t the worst scene loss was 0.0167 with the shield and 0.0011 without it. Together with the fine-tuned-model result, these contrasts show that the confidence shield reduces some unconstrained-selection errors without guaranteeing a visible-relative gain.

The fine-tuned checkpoint switched to fusion at a rate similar to the zero-shot evaluators, so the sign reversal was not explained by a different switching budget. Its calibration error was about five times larger (0.048 versus 0.009–0.010). Among the 70 images on which it selected fusion, 41.4% degraded and only 25.7% improved in per-image AP50; the corresponding degradation fractions were 20.9% and 9.1% for the zero-shot evaluators. The fine-tuned model produced 2 positive and 5 negative scenes. Benjamini–Hochberg correction across the three external contrasts gave $q = 0.043$, 0.043, and 0.119, respectively. This supports a calibration-related boundary hypothesis, but the current data cannot assign causality.

4.6. The shield is a risk-control mechanism

Without the shield, the policy selected fused candidates on most LLVIP and DroneVehicle images and occasionally produced large confidence gains that did not translate into AP. The 1.5 margin reduced switches and lowered the worst AP50 loss from 0.0584 to 0.0040. On DroneVehicle, however, the shield also reduced the mean gain from 0.1040 to 0.0272. This trade-off is the central quantitative meaning of risk control in this study.

Table 8: Diagnostics for the external evaluator boundary. Decision fractions are visible/fused over 300 images. ECE is the detection-level expected calibration error with ten fixed confidence bins. Switched outcomes are improved/unchanged/degraded per-image AP50 among images assigned to a fused candidate. Scenes are positive/negative CS-ECF-minus-visible groups.

Evaluator	Visible/fused	ECE (visible/CS)	Switched I/U/D	Scenes +/−
YOLOv9t, zero-shot	77.7%/22.3%	0.009/0.009	46.3%/32.8%/20.9%	5/2
YOLOv10s, zero-shot	81.7%/18.3%	0.010/0.012	38.2%/52.7%/9.1%	4/3
YOLOv9t, LLVIP-trained	76.7%/23.3%	0.048/0.046	25.7%/32.9%/41.4%	2/5

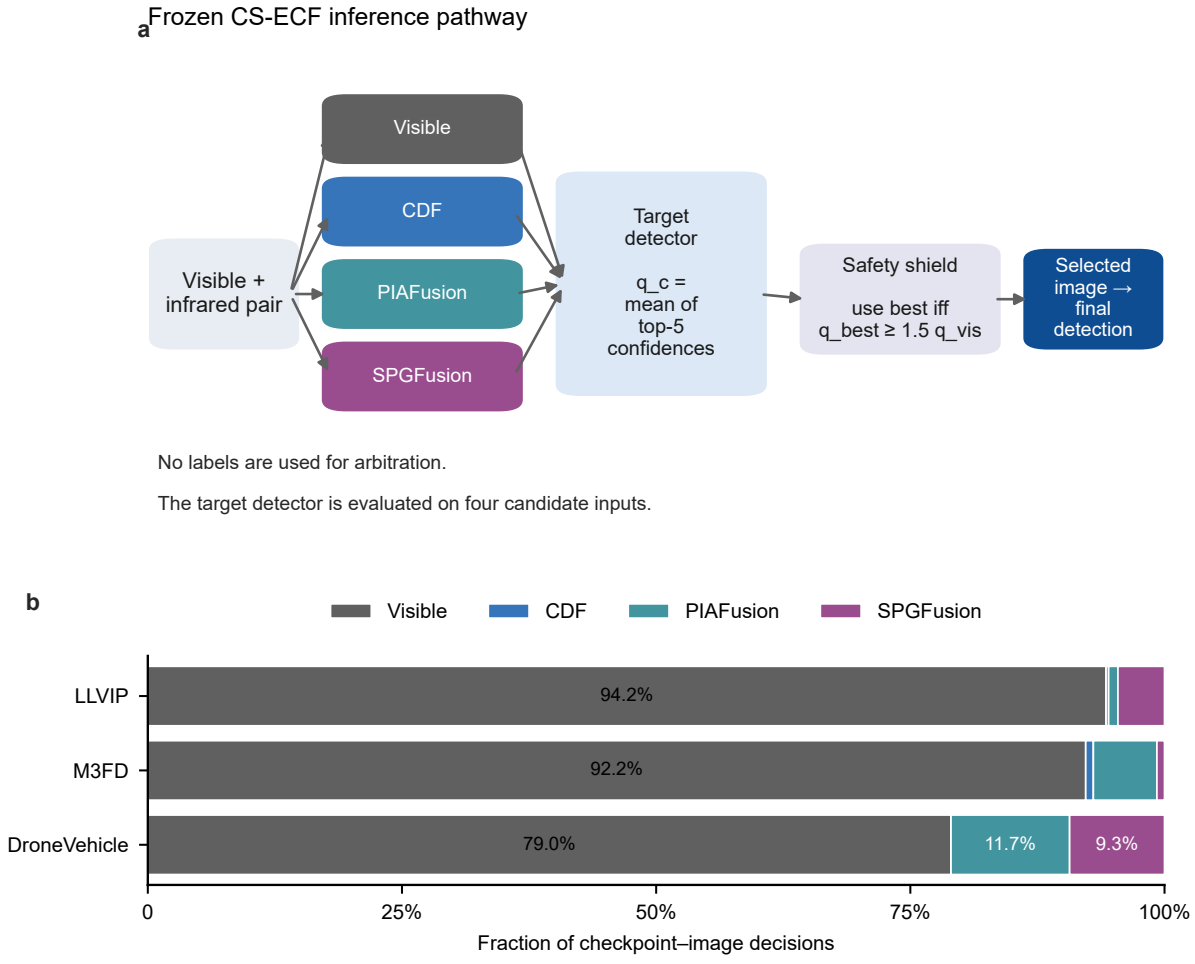


Figure 2: **Confidence-shielded evaluator-conditioned fusion.** **a**, The visible–infrared pair is passed to visible, CDF, PIAFusion, and SPGFusion candidates. The target detector computes its top-five mean confidence for each candidate. A fused candidate is selected only when its score is at least 1.5 times the visible score; otherwise the visible input is retained. No labels are used for arbitration. **b**, Composition of the frozen decisions across all checkpoint–image trials. Source data are provided in `figure2_source.csv`.

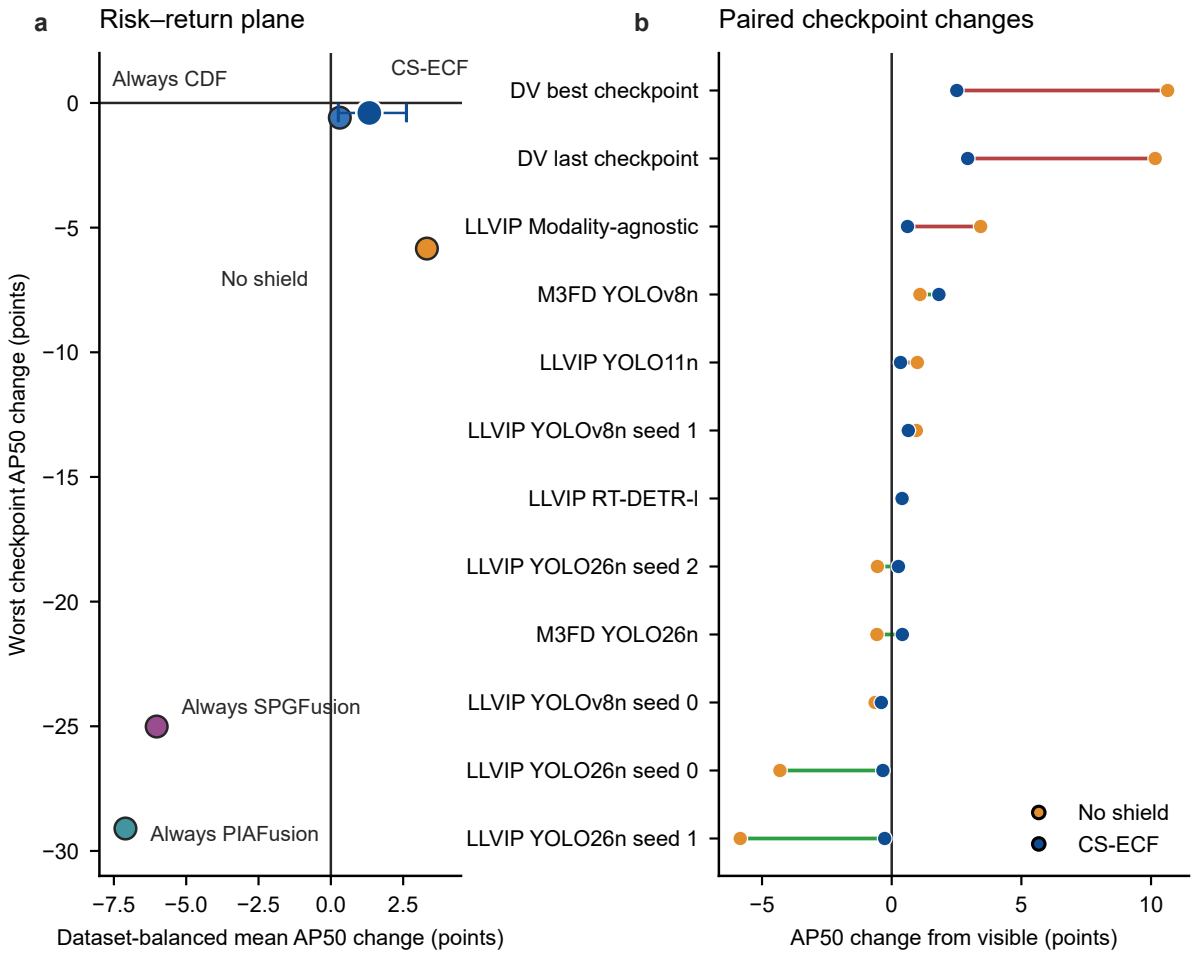


Figure 3: **Risk–return trade-off of fusion-input policies.** **a**, Dataset-balanced mean AP50 change versus worst-checkpoint AP50 change. The CS-ECF horizontal error bar is the training-run-by-group hierarchical 95% interval. **b**, Paired checkpoint changes under the unshielded selector and CS-ECF. The shield sacrifices part of the DroneVehicle mean gain to reduce the worst LLVIP loss. Source data are provided in `figure3_source.csv`.

4.7. Fixed baselines and class concentration

Always CDF improved 10 of 12 checkpoints and had a dataset-balanced gain of 0.0031, but remained negative on both M3FD detectors. Always PIAFusion and Always SPGFusion had negative dataset-balanced gains because their strong DroneVehicle performance did not transfer to the other datasets. In the M3FD six-class evaluation, the CS-ECF gain was concentrated in people: YOLO26n improved by 0.0214 and YOLOv8n by 0.1090, while the other five classes changed by at most 0.0030 in absolute AP50 (Fig. 4). The people class contains 166 of 2,022 test annotations (8.2%), but AP50 weights classes equally. It accounts for 87.7% and 99.6% of the positive class-mean gain for the two detectors, respectively. Image-bootstrap intervals were $[-0.0070, +0.0498]$ for YOLO26n and $[+0.0440, +0.1913]$ for YOLOv8n; only the latter excludes zero. With just two test frame groups, these class intervals are descriptive rather than stable group-level inference. The evidence therefore supports a pedestrian/people-focused interpretation, not uniform six-class improvement.

4.8. Thermal feature association and mask ablation

On LLVIP, infrared top-hat contrast showed a weak positive association with switching (Spearman $\rho = 0.151$, $p = 0.009$), whereas visible luminance was only weakly negative ($\rho = -0.104$, $p = 0.071$). On M3FD, thermal contrast showed a moderate positive association ($\rho = 0.320$, $p < 10^{-5}$). On DroneVehicle, however, both thermal contrast ($\rho = -0.403$) and visible luminance

($\rho = -0.441$) were negatively associated with switching. A high-thermal/low-visible median split did not produce a consistently larger AP gain than the remaining images.

A fixed-input mask ablation gave a smaller but consistent thermal effect on DroneVehicle. Replacing the thermal-validity mask with global detail injection changed the CDF gain from 0.0060 to 0.0057 for the best checkpoint and from 0.0050 to 0.0038 for the last checkpoint. The thermal mask added 0.0007 AP50 on average relative to global injection.

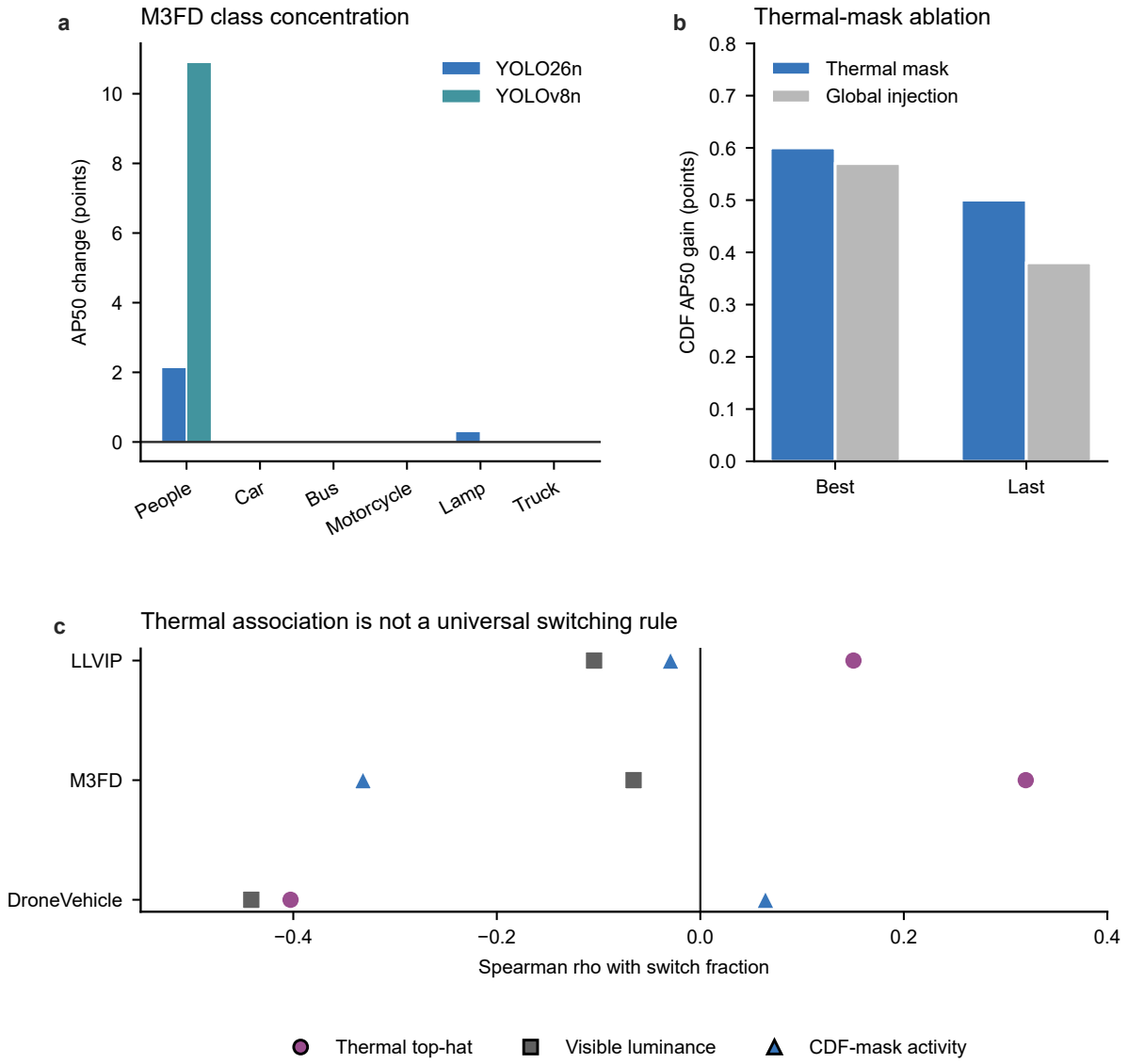


Figure 4: **Thermal constraints and class-specific effects.** **a**, M3FD AP50 changes for six classes. The gain is concentrated in people. **b**, Fixed-input DroneVehicle CDF gain with the thermal-validity mask or global detail injection. **c**, Spearman association between the fused-selection fraction and thermal top-hat contrast, visible luminance, or CDF-mask activity. The association reverses on DroneVehicle, which does not support a universal thermal switching rule. Source data are provided in `figure4_source.csv`.

4.9. Registration sensitivity

We estimated translation residuals for all 300 LLVIP test pairs with ECC on normalized Sobel edges. The median residual was 2.53 pixels. The median test-set person box was 109 pixels wide and 246 pixels high, so the median residual corresponded to 2.3% of median box width and 1.0% of median box height. On the 75 best-aligned pairs (residual at most 1.61 pixels), the mean CS-ECF gain across the eight LLVIP checkpoints was 0.0007, with five positives. On the 75 worst-aligned pairs, the mean gain was 0.0020, with six positives. Most checkpoint-specific

intervals still crossed zero, so these subsets do not identify a separate registration regime. They do show that the observed gain is not confined to the best-aligned quartile. The same residual audit was not available for M3FD or DroneVehicle.

5. Discussion

The central finding is that fusion utility is not an intrinsic property of a fused image alone. A candidate can improve one detector's confidence ordering while harming another's, and a policy that ignores this dependence can select inputs that appear confident but do not improve AP. CS-ECF responds by using the target detector as an input arbiter and requiring a large visible-relative margin before switching.

The shield's value is clearest in the tail. The unshielded selector achieved a higher dataset-balanced average on DroneVehicle, where both PIAFusion and SPGFusion were strong, but it also produced severe LLVIP losses. The 1.5 margin deliberately favors visible fallback under ambiguous evidence. This is consistent with a deployment rule in which a small probability of a large regression is more costly than a missed opportunity for a moderate gain.

The comparison with Always CDF further bounds the use case. When CDF is known to be stable for a fixed evaluator family, the simpler fixed input remains preferable. CS-ECF is relevant when the deployed detector or data domain can move the best fusion family, as observed between pedestrian, mixed-class and vehicle-focused evaluation. Its small balanced advantage should therefore not be read as a demonstrated replacement for CDF everywhere: the interval included zero.

The multiplicity analysis limits the language further. Although the dataset-balanced interval is positive, no individual checkpoint contrast survived Holm correction, and the 9/12 sign test was not significant. The appropriate claim is a small average effect under a prespecified composite estimand, not a collection of proven checkpoint-level wins.

The external evaluators provide a more heterogeneous picture. The two zero-shot checkpoints support a visible-relative gain but not superiority over Always CDF. The LLVIP-fine-tuned YOLOv9t provides the opposite result: CS-ECF falls below visible, while shielding still improves on the unshielded selector. Its detection-level calibration error was about five times larger than those of the zero-shot evaluators, and 41.4% of the images it switched to fusion degraded in per-image AP50. This is consistent with a confidence-calibration failure, but the domain-trained checkpoint also has lower test performance and a larger train-to-test shift. The current design cannot separate those explanations causally.

Four boundaries constrain the interpretation. First, the strongest M3FD effect is class-specific: people detection improves substantially while the remaining classes are nearly unchanged. Second, the DroneVehicle confirmation uses two checkpoints from the same training run; the checkpoints share training data and are correlated. Third, the domain-fine-tuned YOLOv9t demonstrates that an unseen target detector can reverse the sign of the visible-relative contrast. The policy should therefore be validated for each deployed evaluator rather than assumed portable.

A fourth boundary concerns the LLVIP training subset. It was selected lexicographically rather than by scene-stratified sampling, so its scene composition may differ from the full official training partition. The test subset was sampled randomly from all 3,463 non-empty official test pairs.

The thermal stratification adds another limitation. Thermal contrast may increase the likelihood of selecting fused evidence in LLVIP and M3FD, but the association reverses on DroneVehicle. Sensor-level thermal evidence and detector confidence therefore capture different information. Neither should be interpreted as a universal utility score. The appropriate claim is thermal fusion utility under a detector-conditioned decision rule, not thermal dominance.

The cost analysis also matters. SPGFusion dominates latency, and four detector evaluations are not free. CS-ECF is best viewed as a risk-controlled system for settings in which the target

detector is already run and the cost of a catastrophic input choice is high. It is not a low-latency replacement for a single fusion operator. Candidate pruning and early termination were not implemented or claimed; measuring their latency–accuracy trade-off is a separate deployment study.

A practical decision sequence follows from these boundaries:

1. use Always CDF when the detector is fixed and validation shows stable CDF behaviour;
2. consider CS-ECF when the evaluator or domain may change and the target model can be calibrated and revalidated on held-out groups;
3. use the shield when a large regression is more costly than a missed moderate gain;
4. reject the policy for latency-sensitive deployment unless candidate pruning or early termination is evaluated separately;
5. do not deploy the policy after a fine-tuning jump that substantially worsens confidence calibration without rerunning threshold validation.

6. Conclusions

We showed that infrared–visible fusion utility is evaluator-conditioned and proposed CS-ECF, a confidence-shielded input-arbitration policy that falls back to visible unless a detector-supported fusion candidate clears a fixed visible-relative margin. Across three test datasets and 12 checkpoints, the policy improved nine checkpoint point estimates but no individual contrast survived Holm correction; the prespecified dataset-balanced AP50 gain was 0.0133, and the worst observed degradation was reduced from 0.0584 to 0.0040. The dataset-balanced gain over the stronger Always CDF baseline was 0.0102, with a run-cluster by group interval that included zero. The two unused zero-shot evaluators also improved over visible by 0.021–0.025 AP50, without separating from Always CDF. An additional LLVIP-fine-tuned YOLOv9t exposed a counterexample: CS-ECF lost 0.0108 AP50 relative to visible, although it improved the unshielded selector by 0.0153. The evidence supports bounded, evaluator-dependent risk control for pedestrian-centric and mixed infrared–visible detection. It does not support uniform six-class improvement, universal portability across evaluators, independent replication of the DroneVehicle result from one training run, or dominance over Always CDF. The additional cost of SPGFusion and four detector evaluations must also be considered in deployment.

CRediT authorship contribution statement

Jianyong Zhao: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Zigao Wu:** Supervision, Writing – review & editing. **Mengfei Ma:** Investigation, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Data availability

The underlying datasets are publicly available. LLVIP is distributed at `https://github.com/bupt-ai-cz/LLVIP`. M3FD is distributed with the TarDAL benchmark at `https://github.com/JinyuanLiu-CV/TarDAL`. DroneVehicle is distributed through `https://github.com/VisDrone/DroneVehicle` and was obtained from the public archive `https://huggingface.co/datasets/McCheng/DroneVehicle`. Each dataset is used under the licence stated by its distributor. The fixed DroneVehicle test-group manifest, source-archive size and SHA-256, CS-ECF per-image score files, source-data tables, and scripts that regenerate the reported intervals are included in the public analysis repository described below.

Code availability

The code for reproducing the experiments is available at `https://github.com/lessThanbetter/ssgf` under an MIT licence. The version-specific archival release is available at `https://doi.org/10.5281/zenodo.23721458`. The repository and archive contain the fusion and arbitration code, detector-evaluation scripts, fixed split and test manifests, per-image score files, result tables, figure source data, and the numerical checks used to generate the reported intervals.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to support language editing, data-analysis code development, figure preparation, and the organization of numerical results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The detector models, numerical results, and reported conclusions were verified by the authors.

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